

SEARCY, AR, USA

WMO: 722188

Lat: 35.212N

Lon: 91.737W

Elev: 264

StdP: 14.56

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53985

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.7	20.6	4.0	6.8	23.5	9.0	8.8	24.5	19.8	50.2	18.3	45.2	4.8	10	0.421

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	98.2	77.6	95.5	78.1	93.2	77.4	81.6	92.5	80.4	91.0	79.4	89.7	7.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.9	151.7	86.4	77.8	145.9	85.4	76.7	140.6	84.3	45.3	92.5	44.1	91.4	43.0	89.5	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.2	16.3	14.2	DB	9.6	102.2	4.6	5.1	6.3	105.9	3.6	108.8	1.0	111.7	-2.4	115.4
			WB	8.4	83.5	4.3	1.0	5.3	84.2	2.8	84.8	0.4	85.4	-2.7	86.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.3	39.0	43.3	51.9	61.2	69.8	79.1	81.2	80.3	74.4	62.4	50.1	42.1
	DBStd	17.06	10.47	10.81	9.88	7.61	7.14	4.51	4.61	5.06	6.00	8.49	9.01	9.36
	HDD50	1113	368	243	96	9	1	0	0	0	0	8	107	281
	HDD65	3352	805	610	418	158	41	0	0	0	4	154	452	710
	CDD50	5246	28	55	154	344	616	872	966	939	733	392	111	36
	CDD65	2009	0	2	10	44	190	422	501	474	286	73	6	1
	CDH74	20766	0	12	100	410	1705	4567	5566	4990	2655	718	42	1
	CDH80	9637	0	0	8	65	590	2227	2838	2493	1195	217	4	0
Wind	WSAvg	5.7	6.5	7.1	7.5	6.9	5.5	5.0	4.4	4.5	4.7	4.8	5.3	6.1
Precipitation	PrecAvg	49.2	4.0	3.7	5.1	5.0	5.1	2.9	3.5	2.6	3.5	4.4	4.3	4.8
	PrecMax	77.2	7.3	8.0	9.9	14.1	11.3	6.4	7.9	5.5	9.6	14.1	10.2	8.7
	PrecMin	36.8	1.3	1.6	1.6	1.3	1.4	1.1	1.1	0.5	0.3	0.0	0.8	0.7
	PrecStd	10.6	1.9	1.5	2.2	2.9	2.5	1.2	1.7	1.5	2.2	3.5	2.3	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.1	76.0	81.0	85.0	91.7	100.0	102.5	102.6	97.2	90.1	79.8	71.4
		MCWB	62.9	65.8	64.2	70.6	74.2	76.9	76.6	79.8	75.8	72.5	66.5	63.9
	2%	DB	65.2	70.3	77.1	81.9	88.9	96.2	98.2	97.5	94.0	85.7	73.5	66.6
		MCWB	59.9	61.5	64.5	68.0	73.3	78.0	78.1	77.5	76.3	70.8	63.9	62.2
	5%	DB	62.1	66.1	72.4	79.0	86.3	93.4	95.2	94.4	91.1	81.8	69.7	62.4
		MCWB	56.5	59.3	60.9	66.2	72.8	77.5	78.9	77.5	76.4	68.5	62.2	56.9
	10%	DB	57.0	61.7	68.2	75.8	83.4	90.8	92.6	91.8	87.5	77.8	65.2	57.8
		MCWB	50.9	56.0	58.9	64.5	71.3	76.5	78.4	77.2	73.7	66.2	58.1	52.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.9	68.0	68.7	73.7	78.0	82.1	82.8	83.1	80.3	77.0	71.0	67.6
		MCDB	67.6	72.3	78.3	80.9	86.5	93.0	92.4	93.7	91.3	86.0	75.6	69.7
	2%	WB	62.5	64.6	66.4	70.6	76.2	80.3	81.6	81.7	78.7	74.5	66.9	63.4
		MCDB	64.6	68.7	74.0	78.6	84.5	91.3	92.3	92.8	89.5	81.8	71.0	65.8
	5%	WB	57.9	60.7	63.8	68.8	74.7	79.1	80.3	80.5	77.4	71.7	63.3	58.6
		MCDB	60.0	64.6	69.2	75.6	82.7	89.9	90.7	91.0	88.0	77.5	67.6	61.3
	10%	WB	51.5	56.2	61.0	66.8	73.1	77.8	79.2	79.3	76.0	69.3	59.5	53.4
		MCDB	55.1	60.2	66.5	72.9	80.7	88.0	89.2	88.9	84.5	74.3	64.2	55.9
Mean Daily Temperature Range	5% DB	MDBR	19.6	19.4	21.1	22.3	20.9	21.0	20.6	20.7	21.9	23.2	21.8	18.5
		MCDBR	24.6	25.3	26.5	25.9	24.6	24.3	23.4	24.4	25.5	27.3	26.6	23.5
		MCWBR	17.9	18.4	15.5	13.4	11.0	9.0	7.5	9.0	9.7	13.4	16.8	18.0
	5% WB	MCDBR	18.1	20.6	22.4	20.3	20.5	21.1	21.0	21.0	22.3	21.8	20.2	19.8
		MCWBR	16.4	17.0	15.3	12.4	10.4	8.8	8.4	8.6	9.2	13.0	15.9	17.9
Clear-Sky Solar Irradiance	taub	0.321	0.329	0.362	0.399	0.433	0.463	0.493	0.482	0.415	0.360	0.331	0.316	
	taud	2.478	2.454	2.393	2.286	2.253	2.183	2.142	2.152	2.329	2.459	2.494	2.507	
	Ebn at Noon	279	290	288	281	272	262	253	254	267	275	273	275	
	Edh at Noon	27	31	36	42	44	47	49	47	38	30	26	25	
All-Sky Solar Radiation	RadAvg	781	947	1310	1651	1816	2067	1980	1826	1555	1199	852	680	
	RadStd	67	130	107	119	135	143	102	125	120	154	93	74	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page